

Derbyshire Shared Care Pathology Guidelines

Primary Hyperparathyroidism

Driven by hyperfunction of one or more of the parathyroid glands, the hallmark of primary hyperparathyroidism is a circulating level of PTH which is inappropriately high for the level of calcium. Onset of primary hyperparathyroidism most commonly occurs between 40 - 60 years and is more common in women.

Most patients with Primary Hyperparathyroidism are hypercalcaemic but the condition is gradually progressive and intermittent hypercalcaemia can be an early sign. As Vitamin D deficiency can increase PTH levels, correction of vitamin D deficiency is necessary for patients with intermittently raised calcium prior to a diagnosis.

For more information on the aetiology and clinical effects of hyperparathyroidism see page 2 of this guideline.

Investigations and Initial Diagnosis

For the initial diagnosis of Primary Hyperparathyroidism please refer to the Hypercalcaemia Guideline.

If corrected calcium is ≥ 2.5 mmol/L on 2 occasions (2 months apart) **with** renal stones / osteoporosis or significant clinical symptoms indicative of hyperparathyroidism, please request paired calcium (yellow top tube) and PTH (purple top tube).
(NICE NG132)

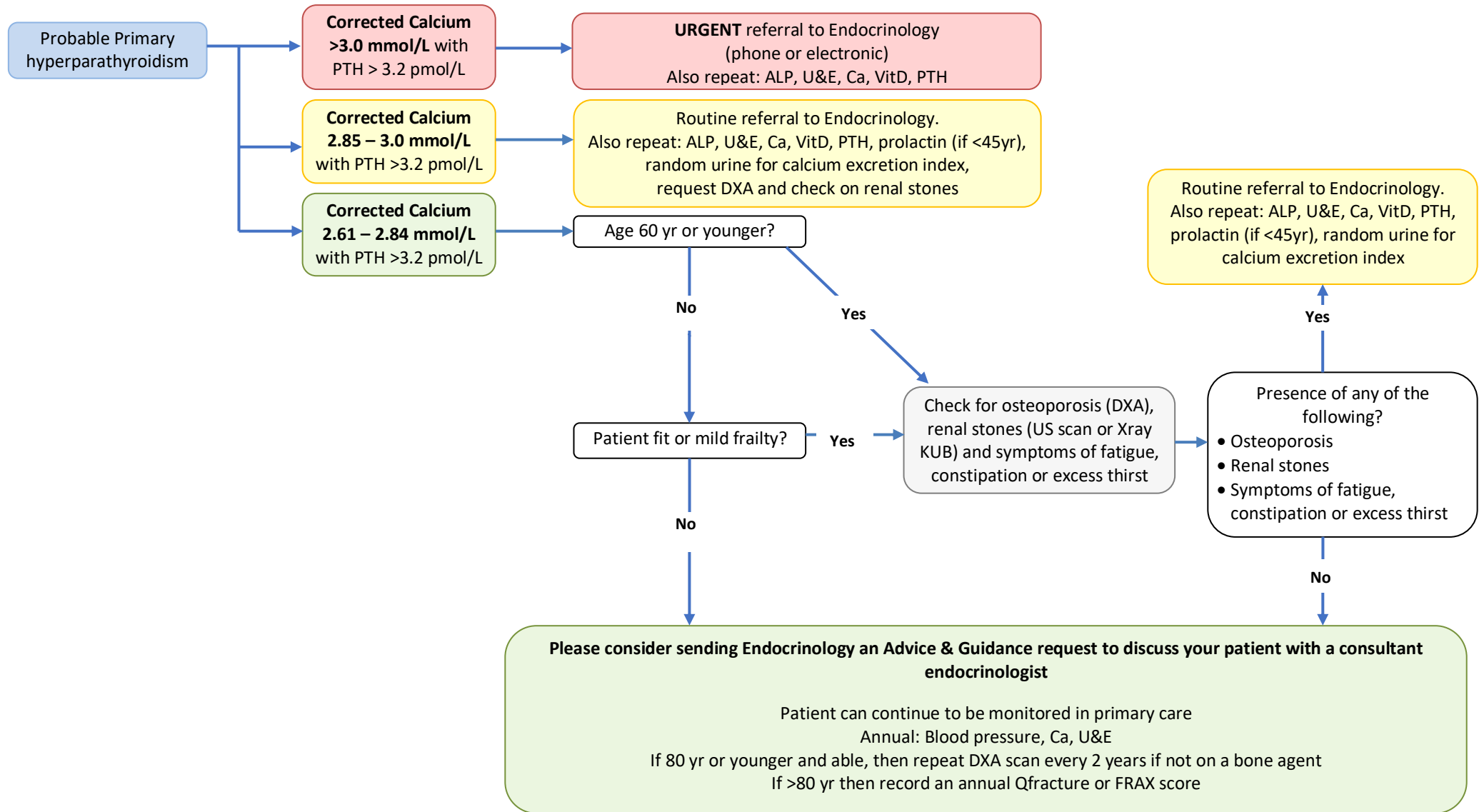
When to Refer

These links are for the flow charts on the following pages for investigation of:

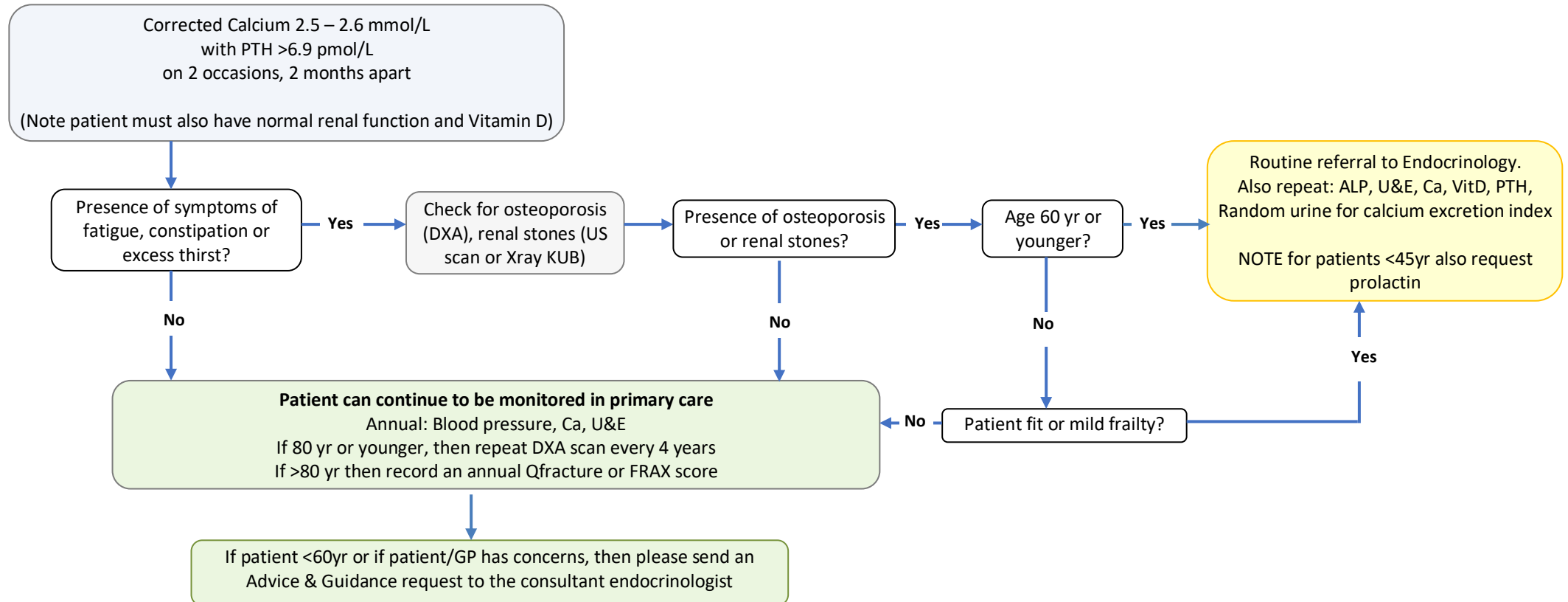
[Probable Primary Hyperparathyroidism](#) (page 2)

[Possible Normocalcaemic Primary Hyperparathyroidism](#) (page 3)

Investigation of probable primary hyperparathyroidism



Investigation of possible normocalcaemic primary hyperparathyroidism



In order for the outpatient appointment to readily identify the patient's needs and treatment please measure the following:

- U&E's
- Bone Profile (Ca, Phos, ALP, Alb)
- PTH
- Serum 25-Hydroxy Vitamin D
- To exclude a genetic condition such as FHH or MEN) also measure:
 - Calcium excretion index - random urine sample (please provide a plain plastic universal container) to be performed with the above blood tests. (Patient should collect the urine sample as close to having bloods taken as possible).
 - Prolactin if the patient is less than 45 years

If calcium 2.61 – 2.84 mmol/L and not requiring an Endocrinology Clinic Referral:

In Primary Care Monitor:

- Serum calcium and renal function — every 12 months
- Blood pressure — every 12 months
- In patients less than 80 years old request a DEXA scan every 2-5 years, as per specialist advice and depending on existing bisphosphonate use (for those on bisphosphonates, every 5 years is appropriate)
- In patients over 80 years assess fracture risk (FRAX) without a DEXA
- Seek advice from Endocrinology if corrected calcium is trending upwards or symptoms develop or end organ damage (renal/bone) is developing or difficult to manage

Useful Background Information

Aetiology of Primary Hypercalcaemia

Over 80% of primary hyperparathyroidism cases are caused by a single parathyroid adenoma of the parathyroid gland.

Other causes include:

- Multiple adenomas (5%)
- Diffuse hyperplasia (12%) - common in lithium treated patients
- Carcinoma (2%)
- Ectopic PTH-secreting tumour (very rare)

In younger patients (<40 yr) with primary hyperparathyroidism please consider a familial cause. Multiple adenomas or hyperplasia also suggests possible genetic aetiology. For example, co-existence of hypercalcaemia with hypertension and neurofibroma is suggestive of Multiple Endocrine Neoplasia type 2.

Clinical effects of Hyperparathyroidism

The clinical features of hyperparathyroidism tend to be non-specific and are related to the aetiology of the hyperparathyroidism and the level of serum calcium.

Most cases are now picked up incidentally during blood investigations and many patients are asymptomatic.

Features may include:

- Polyuria, thirst, weakness, vomiting

- Fracture risk:
 - A study undertaken in 2000 in Denmark (Ref 1) looked at 674 patients with primary hyperparathyroidism, showing that those with primary hyperparathyroidism were at increased risk of vertebral, distal lower limb and forearm fractures before surgery, but not after, compared with a control group (relative risk, RR 1.8 before surgery and 1.0 after surgery)
- Renal calculi in 10-15% of patients
- Bone resorption leading to bone pain or pathological fracture:
 - the main affected areas are where bone turnover is most active such as in the vertebrae, phalanges, ends of long bones and the skull
 - In advanced disease bone resorption leads to formation of bone cysts or brown tumours, with deformities leading to fracture or bending of bones and/or a classical 'pepper-pot skull' seen on plain XR.
 - These changes are now rare as cases tend to be diagnosed at an earlier stage
- Other features of metastatic calcification may be present such as nephrocalcinosis and corneal calcification
- Rarely, primary hyperparathyroidism may present as acute pancreatitis

Contacts

Derby

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
Endocrinology Advice (urgent)	07879 115507 (9am – 5pm, Mon – Fri)
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
Endocrinology Advice (urgent)	01246 513104
Endocrinology advice (non-urgent)	Choose 'Advice and Guidance' from NHS e-Referrals

Burton

Duty Biochemist	01283 566333 ext 4233
Endocrinology advice	Via endocrine secretaries 01283 593259

References

1. Risk of renal stone events in primary hyperparathyroidism before and after parathyroid surgery: controlled retrospective follow up study. Mollerup CL et al. BMJ. 2002 Oct 12; 325(7368): 807.
2. Hyperparathyroidism (primary): diagnosis, assessment and initial management. NICE NG132 (May 2019)

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Dr P Blackwell, Dr R Stanworth, Mrs H Seddon	May 2014	31 st May 2016
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Dr P Blackwell, Dr R Robinson, Dr D Hughes, Dr A Ugur, Mrs H Seddon	Feb 2023	28 th Feb 2025